

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name: Aayam Vashney

Enrollment No. NJG 241475

Jaypee Institute of Information Technology, Noida

Test 1 Examination, 2024

B. Tech, I Semester

Course Title : Physics I

Maximum Time: 1 Hour

Course Code: 15B11PH111

Maximum Marks : 20

CO1	Recall the basic principles of physics related to optics, relativity, radiation, atomic physics and quantum mechanics.
CO2	Illustrate the various physical phenomena with interpretation based on the mathematical expressions involved.
CO3	Apply the concepts/principles to solve the problems related to the wave nature of light, relativity, radiation, atomic physics, and quantum mechanics.
CO4	Analyze and examine the solutions of the problems using physical and mathematical concepts involved.

Note: Attempt all the questions.

- Q1. a) What is the difference between the Fraunhofer and Fresnel class of diffraction?
b) What is the role of the half silvered mirror used in Michelson Interferometer?
c) Will we get the interference pattern in YDSE if we replace the slits with two single filament light bulbs? Justify.
d) Why are the conditions for maxima of the interference pattern in thin films complementary for reflected and transmitted light?
e) What is the color of a very thin film when white light is incident on it? Explain.
[CO1(Remembering), $1 \times 5 = 5$ marks]
- Q2. a) In a Fresnel biprism experiment, source is at 20 cm away from the biprism and 100 cm from the screen. Refractive index of the material of the biprism is 1.5 and the angle is 1° . Calculate the wavelength of incident light if the fringe width is 0.1 mm.
b) Consider a film of refractive index $\mu = 1.38$ and thickness 9×10^{-6} cm. Calculate the wavelength for which the film is non-reflecting.
c) Calculate the resultant amplitude and intensity of superposition of two waves given as $y_1 = 2 \cos(\omega t - \pi/2)$ and $y_2 = 5 \cos(\omega t - 3\pi/2)$.
[CO2 (Understanding), $2 \times 3 = 6$ marks]
- Q3. a) An equiconvex lens of radius 100 cm and refractive index $\mu = 1.4$ is placed on top of another equiconvex lens of radius 150 cm and refractive index $\mu = 1.6$. A monochromatic light of wavelength 500 nm is incident normally on the system. Calculate the radius of the 16th ring if there is a) air b) oil ($\mu = 1.5$) between the lens.
b) A hair of radius 1 mm is placed between two glass plates creating a wedge shaped air film. The distance of the hair from the edge of the wedge is 10 cm. Calculate the distance of 20th dark fringe from the edge of the wedge if the wavelength of incident light is 100 nm. Assume normal incidence.
[CO3 (Applying), $2 \times 3 = 6$ marks]
- Q4. In a Michelson interferometer experiment immersed in water ($\mu = 1.33$), the incident light consists of two wavelengths 600 nm and 590 nm. Calculate the distance by which the movable mirror is moved such that the fringe pattern a) disappears and b) reappears.
[CO4 (Analyzing), $1 \times 3 = 3$ marks]

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Aayon Vanshney

Enrollment No. 9929010159

Jaypee Institute of Information Technology, Noida

T2 Examination, Odd Semester 2024

B.Tech. I Semester

Course Title: Physics - 1

Maximum Time: 1 Hour

Course Code: 15B11PH111

Maximum Marks: 20

CO1	Recall the basic principles of Physics related to optics, relativity, radiation, atomic physics and quantum mechanics.
CO2	Illustrate the various physical phenomena with interpretation based on mathematical expressions involved.
CO3	Apply the concepts/principles to solve the problems related to wave nature of light, relativity, radiation, atomic physics and quantum mechanics.
CO4	Analyze and examine the solution of the problems using physical and mathematical concepts involved.

Attempt All questions.

- Q1. (a) Two persons moving with velocity 2×10^6 m/s notices a light pulse approaching them. While one suggests the speed of the pulse to be 3×10^8 m/s the other one suggests that the speed is higher than 3×10^8 m/s. Who is correct in accordance with the postulates of the special theory of relativity?
- (b) What is inertial frame of reference? Is there only one inertial frame?
- (c) What will be the polarising angle for the air-glass interface and also for the air-water interface? Refractive indices for glass and water are 1.5 and 1.33 respectively.
- (d) What is the effect of increasing number of lines in grating on the intensity of secondary ^{maxima} minima?
- (e) Write the expression for 3-slit diffraction pattern and identify the interference and diffraction terms.

[CO1 (Remembering), $1 \times 5 = 5$ marks]

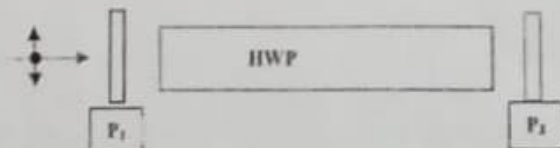
- Q2. (a) Light, which is a mixture of two wavelengths $\lambda_1 = 500$ nm and $\lambda_2 = 540$ nm is incident normally on a plane transmission grating having 2000 lines/mm. A lens of focal length 1 meter is used to observe the diffraction pattern. Calculate the linear separation of two colours in the first order spectrum.
- (b) Find the ratio of intensities of the first and third secondary maxima to that of the central maximum of a single slit diffraction pattern.
- (c) The electric field components of a plane electromagnetic wave are

$$E_x = 2E_0 \cos(\omega t - kz + \phi); E_y = 2E_0 \sin(\omega t - kz)$$

What is the state of polarization for $\phi = \pi/4$? Draw a suitable diagram to show it.

[CO2 (Understanding), $2 \times 3 = 6$ marks]

- Q3. (a) In a double slit diffraction set up, the width of opaque and transparent part are 0.3 mm and 0.1 mm respectively. If a light of wavelength $\lambda = 600$ nm incident on the slits uniformly, find the angular separation between the first and the second diffraction minima. Draw the intensity pattern.
- (b) A HWP is introduced between two crossed polaroids P_1 and P_2 . The optic axis of the HWP makes an angle 15° with the pass axis of P_1 . If the unpolarized beam of light of intensity I_0 is normally incident on P_1 , find the intensities after P_1 , HWP and P_2 respectively.



[CO3 (Applying), $3 \times 2 = 6$ marks]

- Q4. You are given a polariser, a Nicol prism, a HWP, and a QWP. Using above, how can you differentiate between i) an elliptically polarized light and a partially polarized light, ii) a circularly polarized light and unpolarized light?

[CO4 (Analysing), $3 \times 1 = 3$ marks]

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name: Asyan Varshney

Enrollment No. 992491039154

Jaypee Institute of Information Technology, Noida

End Term Examination, Odd 2024

B.Tech. I Semester

Course Title: Physics-I

Maximum Time: 2 hrs

Course Code: 15B11PH111

Maximum Marks: 35

CO1	Recall the basic principles of Physics related to optics, relativity, radiation, atomic physics and quantum mechanics.
CO2	Illustrate the various phenomena with interpretation based on mathematical expressions involved.
CO3	Apply the concepts/principles to solve the problems related to wave-nature of light, relativity, quantum mechanics and atomic physics
CO4	Analyse and examine the solution of the problem using physical and mathematical concepts

Note: Read the instructions carefully. Attempt all questions. Notations and symbols have usual meaning.

Q1	Attempt any 5 questions from Q1(a) to Q1(f)
	a) Explain the principal quantum number (n) and magnetic quantum number (m_l) for an atomic electron.
	b) State Wein's displacement law with its expression.
	c) A particle of mass m and charge q is accelerated by voltage V . Write the expression for its de-Broglie wavelength. Is it possible to accelerate this particle so that it reaches speed of light? Explain briefly.
	d) Draw the energy density curve for a black body indicating the ultra-violet catastrophe in it.
	e) Write any two properties of an acceptable wavefunction of a quantum system.
Q2-8	f) Is it possible for a photon to have its total energy equal to its rest mass energy? Justify, briefly. [CO1 (Remembering), 5 x 1 = 5 marks]
	Attempt any 5 questions from Q2 to Q7. Q8 is compulsory.
	2) What is space quantization? Estimate the possible orientations of orbital angular momentum $\vec{L}$ for an atomic electron in the state of orbital quantum number $l=1$.
	3) Indicate the group velocity (V_g) and the phase velocity (V_p) in schematic diagram of a wave packet. Obtain the relation $V_g = V_p \frac{dV_p}{d\lambda}$
	4) Find the normalized wavefunction for a particle of mass m , confined in a 1-D box ($0 \leq x \leq L$) of width L . Draw its wavefunction and probability density, for the first excited state.
	5) Show that Wein's law and Rayleigh Jeans law are special cases for Planck's radiation law.
	6) Using $E^2 = p^2 c^2 + m_0^2 c^4$, where E , p and m_0 are total energy, momentum and rest mass of particle respectively, deduce the classical expression for kinetic energy for $v \ll c$.
	7) Write the time dependent and time independent Schrödinger wave equations. Find the energy difference (in terms of ground state energy E_0) between 1 st excited state and 3 rd excited state for a particle in 1-D box.
	8) Write any two differences between interference pattern obtained in the Newton's ring and Michelson interferometer. [CO2 (Understanding), 6 x 2 = 12 marks]
Q9-13	9) A single slit of width 0.2 mm is illuminated by light of wavelength 600 nm and the Fraunhofer diffraction pattern is observed through a lens of focal length 50 cm. Find the linear width of central maxima. Draw the corresponding intensity pattern up to 2 nd order minima on both sides, showing its angular positions.
	10) An impure sample of sugar (100 gm) is dissolved in 1 Litre of water. The solution produces the optical rotation of 10° using the polarimeter tube of length 20 cm. If the specific rotation of pure sugar is $66^\circ \text{ dm}^{-1} (\text{gm/cc})^{-1}$, find the % purity of sugar sample. Also, find the optical rotation if the concentration of sugar sample is doubled.
	11) Two spaceships A and B moving in same direction, away from Earth are observed to have velocity $0.6c$ and $0.9c$ with respect to (w.r.t.) Earth. Spaceship A launches a missile towards B, at velocity $0.4c$ w.r.t itself. What is the velocity of missile w.r.t. Earth and w.r.t. spaceship B.
	12) Find the expectation values $\langle p \rangle$ and $\langle E \rangle$ for wavefunction $\psi = A e^{i(kx - \omega t)}$, corresponding to a free particle.
	13) An electron beam of energy 300 eV is passed through a circular hole of radius 1 micrometre. Find the corresponding uncertainty in its momentum and its angle of emergence. [CO3 (Applying), 5 x 3 = 15 marks]
Q14	An electron of energy $E = 1 \text{ eV}$ strikes 1-D rectangular barrier of height $V_0 = 10 \text{ eV}$ and the barrier width $L = 1 \text{ \AA}$. Analyse the effect on the transmission coefficient for the following cases: Case I: Doubling the electron energy E and keeping width L unchanged Case II: Doubling the width L and keeping electron energy E unchanged [CO4 (Analysing), 1 x 3 = 3 marks]

Constants: $m_e = 9.1 \times 10^{-31} \text{ kg}$; $h = 6.6 \times 10^{-34} \text{ J-s}$; $e = 1.6 \times 10^{-19} \text{ C}$; $c = 3 \times 10^8 \text{ m/s}$